

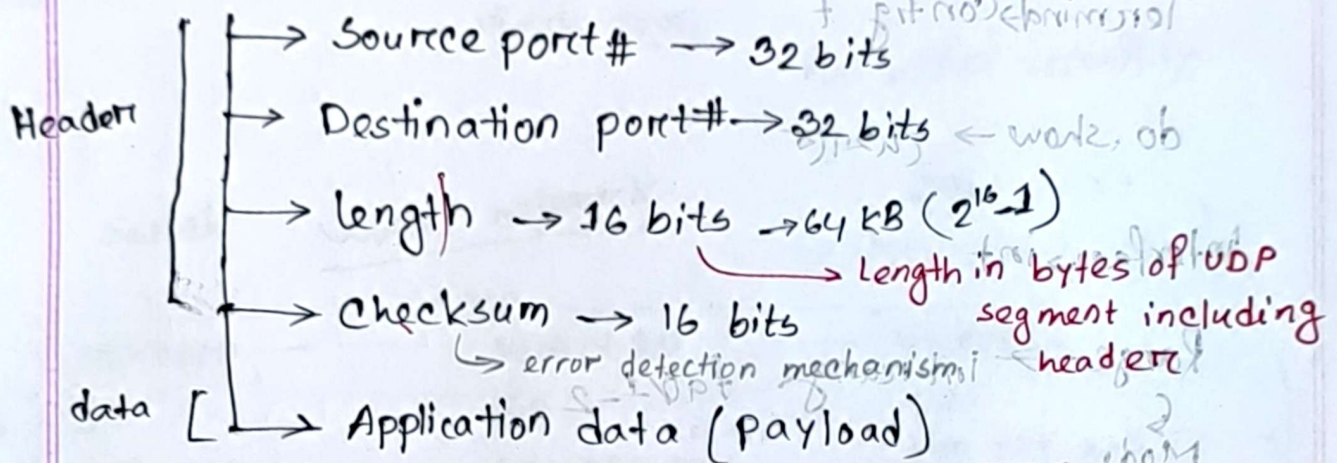
UDP Segment Header

Chapter - 3 [TCP, UDP]

UDP Segment Header

UDP → No service

5 Parts [32 bits]



Q. Describe a UDP segment.

↳ Description of the 5 parts

Internet checksum: an example

Example: add two 16 bit integers

NB: When adding numbers a carry needs from the most significant bit need to be added to be re

1st complement

Wrap around

carry

Sum

invert

```

  1 1 1 0 0 1 1 0 0 1 1 0 0 1 1 0
  1 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
  -----
  1 1 0 1 1 1 0 1 1 1 0 1 1 1 0 1
  1 1 0 1 1 1 0 1 1 1 0 1 1 1 0 1
  -----
  1 0 1 1 1 0 1 1 1 0 1 1 1 1 0 0
  1 0 1 1 1 0 1 1 1 0 1 1 1 1 0 0
  -----
  0 1 0 0 0 1 0 0 0 1 0 0 0 0 1 1
  
```

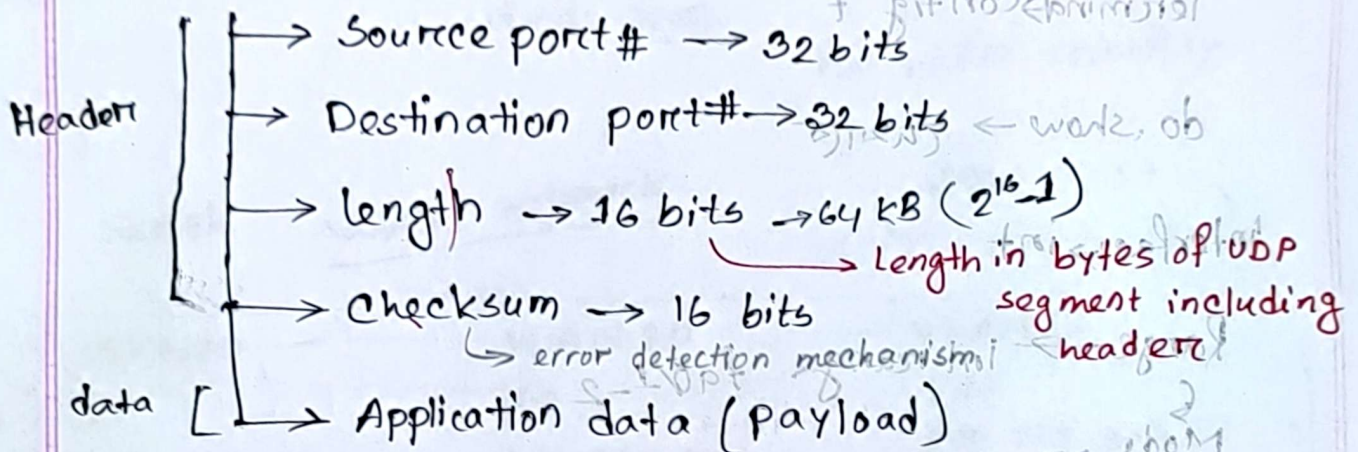
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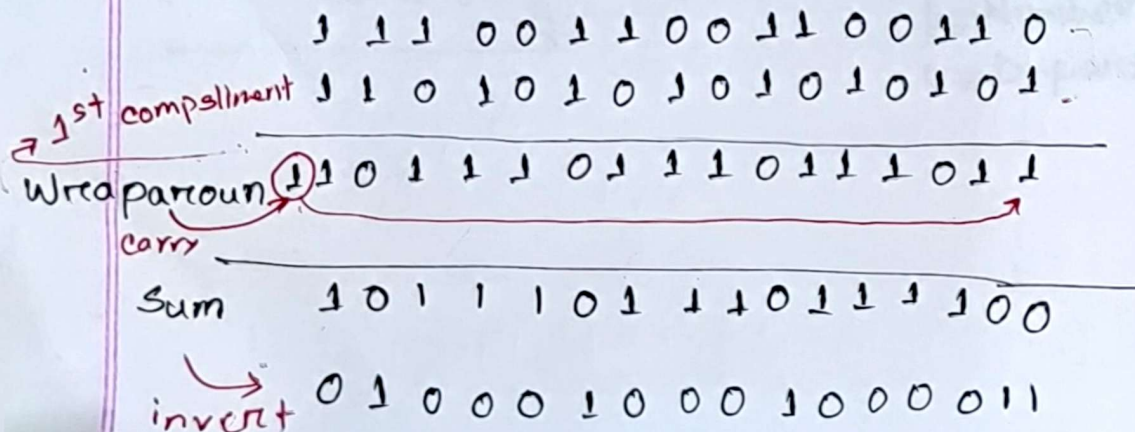
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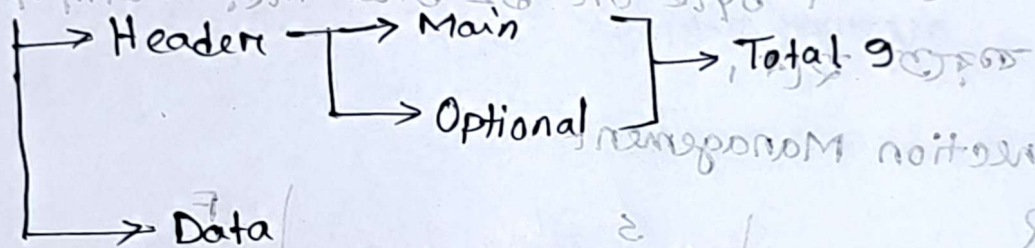


Process

- Normal binary addition
- Carry wraparound $\rightarrow$ Sum
- Invert

Next expected (ACK) = $seq + 1 = 100$

TCP segment structure



Receive window $\Rightarrow$ Amount of data can be received

$\rightarrow$ Flow control $\rightarrow$ bytes receiver willing to accept

ACKnowledgment number (ACK) $\rightarrow$ sequence of next expected byte

connection set এর সময়
sequence number set করতে হবে

200 byte

1st data sequence number = 500
 $\text{Data size} + 1^{\text{st}} - 1$

Last data sequence = $500 + 200 - 1$
 $= 699$

Next expected (ACK) = $699 + 1 = 700 \rightarrow \text{Last} + 1$

• Header length bit এ মা বজানো হবে

তাকে 4 byte or 32 bits দিয়ে multiply
 করতে হবে।

• Connection Management

R

- RST

S

- SYN

- Connection

set

→ connection

set করার সময়

set করে হয়

F

- FIN

- connect finist

→ connection finish

* Acknowledgement number (ACK)

→ sequence of next expected
 byte

U → Urgent data pointer. valid if u bit is set

④ TCP round trip time, timeout

$$\text{Estimated RTT} = \underbrace{(1-\alpha)}_{\text{New}} * \overset{\text{Previous}}{\text{Estimated RTT}} + \alpha * \text{Sample RTT}$$

Typical value of $\alpha = 0.125$

- Timeout interval: Estimated RTT plus "safety margin"
- Large variation in Estimated RTT: wants a large safety margin

~~$$\text{Timeout Interval} = \text{Estimated RTT} + 4 *$$~~

$$\text{Timeout Interval} = \text{Estimated RTT} + 4 * \text{Dev RTT}$$

→ safety margin

- Dev RTT: EWMA of Sample RTT deviation from Estimated RTT

$$\text{Dev RTT} = (1-\beta) * \text{Dev RTT} + \beta * (\text{Sample RTT} - \text{Estimated RTT})$$

Typically $\beta = 0.25$